

The line will have the same y intercept *at the origin of the graph.*

Q5(a)(i)

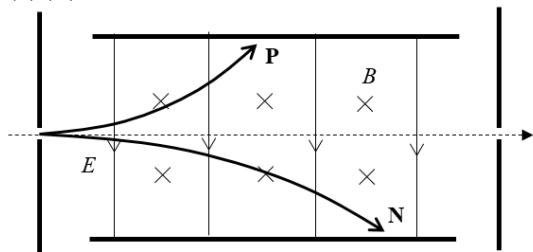
For zero deflection, net force on ion = 0

$\therefore$ electric force = magnetic force

$$qE = Bqv$$

$$\therefore v = \frac{E}{B} = \frac{1210}{0.015} = \mathbf{8.1 \times 10^4 \text{ m s}^{-1}}$$

5(a)(ii)



5(b)(i)

For circular motion,

Centripetal force = magnetic force

$$\frac{mv^2}{r} = Bqv$$

$$\therefore r = \frac{mv}{Bq}$$

The positive ions are deflected downward into the lower circular path of radius = 1.2 cm.

$$\therefore r = \frac{mv}{Bq}$$

$$1.2 \times 10^{-2} = \frac{m \times 8.1 \times 10^4}{1.6 \times q}$$

$$\therefore \frac{q}{m} = \mathbf{4.2 \times 10^6 \text{ C kg}^{-1}}$$

The negative ions are deflected upward into the upper circular path of radius = 1.8 cm.

$$\therefore r = \frac{mv}{Bq}$$

$$1.8 \times 10^{-2} = \frac{m \times 8.1 \times 10^4}{1.6 \times q}$$

$$\therefore \frac{q}{m} = \mathbf{2.8 \times 10^6 \text{ C kg}^{-1}}$$

5(b)(ii)

For the singly-charged positive ion,

$$\frac{q}{m} = \frac{(e)}{N_n(u)} = 4.2 \times 10^6$$

where

e = elementary charge = $1.6 \times 10^{-19} \text{ C}$,

N_n = no. of atomic mass units,

u = atomic mass unit = $1.66 \times 10^{-27} \text{ kg}$ $\approx$ mass of a nucleon.

[1]

$$N_n = \frac{1.6 \times 10^{-19}}{(4.2 \times 10^6)(1.66 \times 10^{-27})}$$

$$N_n \approx 23$$

Hence, the positive ions could be **sodium ions** (Na^+).

For the negative ion,

$$\frac{q}{m} = \frac{(e)}{N_n(u)} = 2.8 \times 10^6$$

$$N_n \approx 34$$

Hence, the negative ions could be chlorine ions (Cl^-).

Q6(a)